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MILITARY SPECIFICATION

BOILERS, FLUIDIZED BED, STEAM AND HOT WATER, PACKAGED TYPE (330 TO 51,500 POUNDS PER HOUR THERMAL OUTPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers watertube and firetube multipass fluidized bed combustion packaged steam boilers or hot water generators.

1.2 Classification. The steam boiler or hot water generator shall be of the following types, sizes, and classes as specified (see 6.2).

TYPES

Type I - Steam boiler producing low pressure steam not greater than 15 pounds per square inch gage pressure (psig).
Type II - Steam boiler producing high pressure steam between 16 and 150 psig.
Type III - Steam boiler producing high pressure steam between 151 and 350 psig.
Type IV - Steam boiler producing high pressure steam between 351 and 600 psig.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Office (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 4410

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TYPES - Continued

Type V - Hot water generator for low temperature water not exceeding 160 psig or 250 degrees Fahrenheit (°F).
Type VI - Hot water generator for high temperature water exceeding 160 psig or 250°F.

SIZES

Size 2 - Thermal output capacity between 320,001 and 10,000,000 British thermal units per hour (Btu/hr).
Size 3 - Thermal output capacity between 10,000,001 and 35,000,000 Btu/hr.
Size 4 - Thermal output capacity between 35,000,001 and 125,000,000 Btu/hr.

CLASSES

Class 1 - Heavy oil fired.
Class 2 - Light oil fired.
Class 3 - Gas fired.
Class 4 - Combination heavy oil and gas fired.
Class 5 - Combination light oil and gas fired.
Class 6 - Coal fired.
Class 7 - Coal/oil fired.
Class 8 - Coal/gas fired.
Class 9 - Coal/oil/gas fired.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-B-28685 - Baghouse, Reverse Pulse Jet; For Packaged Boilers Fired on Coal or a Combination of Coal and Gas, Coal and Oil, or Coal-Gas-Oil (2,000,000 to 125,000,000 Btu/hr Thermal Output Capacity).

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STANDARD

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Boiler Water Limits and Steam Purity Recommendations.
Recommended Design Guidelines for Stoker Firing of Bituminous Coals.

(Application for copies should be addressed to the American Boiler Manufacturers Association, 950 N. Glebe Rd., Suite 160, Arlington, VA 22203.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section I - Power Boilers.
Section IV - Heating Boilers.
Section VIII - Pressure Vessels.
ASME CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.
ASME Performance Test Code.
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM D 396 - Fuel Oils.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Factory Mutual Approval Guide.

(Application for copies should be addressed to the Factory Mutual Engineering Corporation, Factory Mutual System, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA ICS 2 - Industrial Control Devices, Controllers, and Assemblies.
NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 85A - Explosion Prevention in Fuel Oil- and Natural Gas-Fired Single Burner Boiler-Furnaces.
NFPA 85H - Fluidized Bed Burner.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 353 - Limit Controls.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The units covered by this specification shall either be a steam boiler or a hot water generator as specified (see 6.2). The steam boiler or hot water generator shall be capable of producing steam or hot water in the pressure or temperature type specified (see 6.2). The steam boiler or hot water generator shall be capable of the thermal output capacity specified (see 6.1.1 and 6.2). The unit shall be furnished with a complete fluidized bed

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combustor fuel system, air movement system, electrical system, and control system. The unit shall be furnished ready for operation when connected to water, fuel, and electrical connections.

3.2 Codes and standards. The steam boiler or hot water generator shall conform to the applicable requirements of Section I, Section IV, and Section VIII of the ASME Boiler and Pressure Vessel Code (hereinafter referred to as the ASME Code), ASME CSD-1, FM, NFPA 70, NFPA 85A, and NFPA 85H. Limit controls, interlock switches and other components shall be in accordance with UL 353 or FM.

3.3 Standard commercial product. The steam boiler or hot water generator shall, as a minimum, be in accordance with the requirements of this specification and may be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the burner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification shall be new.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Design. The steam boiler or hot water generator shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation. The unit shall be designed and constructed in accordance with the applicable requirements of 3.2 and the additional requirements specified herein. The steam boiler or hot water generator shall be designed for the following operating parameters as specified (see 6.2):

a. Design pressure (psig).
b. Operating pressure at packaged unit outlet (psig).
c. Design temperature (^{o}F).
d. Operating temperature at packaged unit outlet (^{o}F).
e. Maximum continuous thermal output (Btu/hr).
f. Feedwater temperature (^{o}F).

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g. Maximum, minimum, and average anticipated ambient air temperatures (^{o}F).
h. Site elevation above sea level (feet).

3.7.1 Fuel. The steam boiler or hot water generator shall be fired on the fuel(s) specified (see 6.2). The start-up fuel shall either be natural gas or propane as specified (see 6.2). Performance requirements shall be based on the following fuel characteristics.

3.7.1.1 Coal fuel characteristics. The description of the coal and testing procedures used to establish the coal characteristics shall be in accordance with Appendix A of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals. The fluidized bed combustor shall be designed for coal of the following characteristics as specified (see 6.2):

a. Proximate analysis - as received (weight percent)

(1) Volatile matter
(2) Fixed carbon
(3) Ash
(4) Moisture (total)
(5) Equilibrium moisture

b. Higher heating value

c. Ash analysis (percent by weight)

d. Ash fusion temperature in reducing and oxidizing environments

(1) Hemispherical (Height equals one-half width)

e. Free swelling index

f. Coal source

(1) State
(2) Mine

g. Coal size distribution

3.7.1.2 Fuel characteristics. Unless otherwise specified (see 6.2), oil fired burners shall be designed and constructed to operate on the fuel oil characteristics described in ASTM D396. When oil burners are required, grade of fuel required for firing and fuel oil supply temperature and pressure shall be as specified (see 6.2). When gas burners are required, fuel gas supply pressure shall be as specified (see 6.2). Unless otherwise specified (see 6.2), gas burners shall be designed and constructed to operate on the following fuel gas characteristics:

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a. Natural gas, (percent by volume, as fired).

(1) Ultimate analysis:

(a) Methane ---------- 90.0
(b) Ethane ---------- 5.0
(c) Nitrogen --------- 5.0
(d) Ash ------------- Nil
(e) Moisture --------- 0.01

(2) Higher heating value - 1,075 British thermal units per cubic foot (Btu/cu ft).

3.7.2 Turndown ratio. Unless otherwise specified (see 6.2), the continuous stable low firing rate shall be not greater than 33 percent of the maximum required continuous high firing rate. The low firing rate shall be attained automatically by the combustion control system during a normal firing cycle without interruption or changing of firing components (see 6.1.1).

3.7.3 Thermal shock. The steam boiler or hot water generator shall be designed to withstand the thermal stresses associated with cold start and emergency shutdown operating conditions.

3.8 Performance characteristics.

3.8.1 Thermal efficiency. When tested in accordance with 4.7.2, the thermal efficiency shall be not less than 80 percent at any firing rate. This efficiency requirement is independent of auxiliary heat recovery units.

3.8.2 Steam purity. When a steam boiler is required (see 3.1), the total dissolved solids (TDS) in the steam shall be in accordance with the ABMA Boiler Water Limits and Steam Purity Recommendations for Watertube Boilers shown in Table I. The moisture content of the steam leaving the steam drum shall be not greater than 0.5 percent for boiler water TDS concentrations of less than 3,000 parts per million. Firetube style boilers shall only meet the steam moisture content requirement.

TABLE I. Boiler water quality and associated steam purity.

Steam Drum Pressure (psig)	Range of Feedwater TDS (parts per million)	Range of Steam TDS (parts per million)
0 - 300	700 - 3,500	0.2 - 1.0
301 - 450	600 - 3,000	0.2 - 1.0
451 - 600	500 - 2,500	0.2 - 1.0

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3.8.3 Thermal heat input rates. When the steam boiler or hot water generator is operating at the maximum specified thermal output, the thermal input to the furnace shall be such that none of the following rates specified herein shall be exceeded.

3.8.3.1 Volume heat input rate. The volume heat input rate (see 6.4), shall be not greater than 80,000 Btu/hr per cu ft of the furnace volume for coal firing.

3.8.3.2 Surface heat input rate. The maximum heat input rate of effective radiant heating surface of the furnaces shall be not greater than 60,000 Btu/hr per square foot (sq ft) (see 6.4).

3.8.3.3 Heat transfer rate. The heat transfer rate shall be determined on the basis of the thermal output divided by the total generator heating surface (see 6.4) and shall be not greater than 8,400 Btu/hr per sq ft.

3.8.4 Exit temperature at stack. Unless otherwise specified (see 6.2), the exit temperature of the flue gas at the stack shall be not less than 325°F without an economizer. When exit temperatures of less that 325°F without an economizer are allowed, the flue gas passages shall be of corrosion resistant materials to prevent potential flue gas condensate corrosion problems.

When operating at minimum load and an economizer is required (see 3.12), the exiting flue gas temperature shall be not less than 300°F. The economizer water temperature at flue gas exit shall not be less than 220°F. When exit flue gas temperatures of less than 300°F are allowed use corrosion resistant materials in the flue gas passages and economizer to prevent potential flue gas condensate corrosion problems.

3.8.5 Pressure and temperature regulation. The combustion control system shall maintain the steam pressure or water temperature, as applicable, at the outlet connection within ±7 percent expressed as a percent of the set point values of the operating control.

3.8.6 Environmental requirements. The emission requirements shall be met at all operating loads, including the maximum required continuous output. The steam boiler or hot water generator shall meet the legal emission requirements required by the local, state, and Federal environmental rules and regulations. The emission requirements for opacity, particulates, nitrogen oxides, sulfur dioxide, and carbon monoxide shall be as specified (see 6.1.1 and 6.2).

3.8.7 Baghouse design data requirements. When specified (see 6.2), the boiler manufacturer shall provide the following information to be used for the design and installation of a flue gas baghouse in accordance with MIL-*B-28685:*

a. Maximum gas flow in standard cubic feet per minute.
b. Required draft at boiler flue gas outlet in inches of water at maximum gas flow conditions.
c. Dimensions of flue gas outlet.
d. Height of flue gas outlet above grade and relative orientation.

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3.9 Construction. Steam boilers and hot water generators shall be constructed for the specified design conditions and performance requirements in accordance with the applicable sections of the ASME Code and the additional requirements specified herein.

3.9.1 Hot water generator. When a hot water generator is furnished (see 3.1), it shall be designed to operate in a flooded condition without a steam space for pressure control. The expansion or pressure regulating vessel, feed water regulating valve, and associated controls are not considered part of this specification when a hot water generator is required (see 6.1.1).

3.9.2 Water pressure drop. Unless otherwise specified (see 6.2), when a hot water generator is furnished (see 3.1), the water pressure drop shall not be greater than 20 psig. Flow restrictions causing temporary pressure drops shall not cause cavitation or flashing of the hot water when the generator is operated at the required temperature and pressure.

3.9.3 Combustor. Unless otherwise specified (see 6.2), the combustor chamber may either be of the bubbling bed or circulating bed design. Combustion surfaces shall either water cooled or covered by abrasion resistant refractory. Multiple bed construction may be used for achieving stable low firing rates or high thermal efficiency. The walls and floor of the combustor shall permit free thermal expansion of the packaged unit without thermal stress damage on any part of the unit. Waterwall headers and supply tubes shall be protected from the product of combustion.

3.9.4 Tubes. Tubes shall be either electric-resistance welded or seamless steel. Firetubes shall be rolled and beaded into place. Not less than one separate tubesheet for each exhaust gas pass shall be used to minimize thermal stresses in the firetube design tube bundle. Spinners, turbulators and other such devices shall not be permitted in the tubes. Tubes may be metal sprayed, finned, or covered with refractory to prevent erosion damage.

3.9.5 Refractories. Refractories shall withstand the temperature existing under maximum load conditions. The refractories shall be formed in place or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment. Refractory shall be erosion resistant in combustor locations where high bed velocities exist.

3.9.6 Access doors and observation ports. The steam boiler or hot water generator shall be equipped with access doors for normal field internal inspection, cleaning and maintenance of the furnace fireside. Not less than one observation port shall be provided to permit operators to see the combustor interior and flame pattern. Observation ports shall be fitted either with safety shields to maintain closure unless held open by the operator, or with air-cooled heat resistant glass closures. Access doors and observation ports shall be gas-tight.

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3.9.7 Baffles. Baffles, as required, shall be arranged to diffuse the gas properly over heating surfaces in order to obtain maximum heat absorption. The baffles shall be held securely in place without being affected by or interfering with the free expansion or contraction of the steam boiler. The baffles shall be watertubes, or of steel plate not less than 0.25 inch thick, or of heat resistant alloy not less than 0.125 inch thick. Tangent tube baffles may be used in lieu of metal plate baffles. Baffles shall withstand the combustor temperatures under maximum load conditions. The partition wall between the furnace and tube bank shall be of membrane wall construction if the remainder of the furnace is of similar construction.

3.9.8 Insulation. All exterior surfaces of the steam boiler or hot water generator, including the floor, but excluding doors, drumheads, hand holes, manholes, vents, and appurtenances, shall be insulated and jacketed. No asbestos material shall be used. The insulation shall be so formed and secured in place as to prevent sagging or displacement during shipment and operation. The insulation shall be covered with a sheet metal jacket made of steel having a thickness of not less than 0.023 inch (24 U.S. Standard Gage) or of nonferrous sheet metal having a thickness of not less than 0.029 inch (22 U.S. Standard Gage). All sections of the steam boiler or hot water generator exposed to the products of combustion shall be backed by refractory, by insulation, or both, to maintain a casing temperature of not greater than 130°F with a surface air velocity of 5 feet per second and an ambient temperature of 80°F while the steam boiler or hot water generator is operating at maximum capacity.

3.9.9 Mechanical dust collector. The collector shall be of the cyclone or impinger design. The combustion gases from the steam boiler or hot water generator shall be vented to the inlet of the collector. An airtight dust receptacle shall be provided as an integral part of the steam boiler or hot water generator. Suitable means for emptying the collected dust particles from the receptacle shall be provided. The collector shell and all sheet metal parts shall be constructed of steel. All joints and seams shall be made airtight. The collector shall not pass more than 2.0 percent by weight of conveyed material to the flue gas exhaust that is 50 microns and larger.

3.9.10 Carbon reinjection. A suitable means for the reinjection of the carbon from the combustion gases to the furnace shall be provided. The sides of the carbon collection chamber shall be at such an angle as to prevent bridging over of the carbon particles. The carbon reinjection system shall be furnished with a nonmechanical seal valve.

3.9.11 Ash handling system. The steam boiler or hot water generator shall be furnished with an underbed ash removal system and collection hopper. A manually controlled ash recycle system shall be provided for reintroduction of the flyash to the furnace. Unless a conveyor system is specified (see 6.2), ash removal from the collection hopper shall be with a manually operated door.

3.9.12 Skid base. The steam boiler or hot water generator shall be mounted on steel saddles or equivalent steel supports fastened securely to a structural steel base designed and constructed to make a complete

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self-contained unit that will require only a flat surface, capable of supporting the weight of the steam boiler or hot water generating unit when filled with water, for a foundation.

3.9.13 Draft fans. The centrifugal draft fans shall be for a balanced system and shall include a forced draft and an induced draft fan system. Oil reservoirs shall be equipped with an oil level sight glass and drain. The draft fans shall provide sufficient air to allow the steam boiler or hot water generator to meet the specified performance requirements. Draft fan casings shall be provided with access doors and drain plugs. Adequate margin shall be included in the fan selection or design for maximum pressure drop, combustor pressure loss, combustion air temperature, and plant elevation. The induced draft fan shall be constructed to withstand the hot combustion gases.

3.9.14 Inner casing. If an inner casing is included in the manufacturer's standard design, it shall be pressure-tight to prevent the products of combustion from coming in contact with the boiler insulation and outer casing. The inner casing may either be the tube wall itself or a separate inner casing having a thickness of not less than 0.135 inch, placed just behind the tubes or the refractory backing. The casing shall be pressurized for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches of water, whichever is less. The pressure drop shall be not greater than 5 inches of water in ten minutes.

3.10 Steam boiler and hot water generator trim. Each steam boiler or hot water generator shall be fully equipped with the trim required under the applicable section of the ASME Code for the type of unit specified herein. The additional appurtenances specified herein shall also be furnished. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The trim, valves, connecting piping and fittings shall be suitable for the design pressure and temperature as specified herein. Limit controls and interlock switches shall be in accordance with the applicable requirements of FM or UL-353. Unless otherwise specified (see 6.2), the trim and appurtenances shall include but not necessarily be limited to the following:

a. High water level alarm for steam boilers.
b. Low water level alarm and a low water level fuel cutoff switch. The switch shall be mounted on a separate bridle with a drain valve.
c. A manual bypass momentary contact push button switch for the low water level cutoff switch of item b shall be located adjacent to the drain valves for the low water cutoff switch.
d. Piping and valves on steam boilers for continuous blowdown. The continuous blowdown valve shall have stainless steel trim and be able to maintain the position of a manual setting.
e. A top drum vent valve with threaded connection and corrosion resistant trim.
f. A chemical feed pipe connection with a diameter of not less than 0.75 inch with a corrosion resistant trimmed stop valve installed on the top and on the feedwater inlet piping.
g. A temperature gage on the fluid outlet. This temperature gage may be one of the four gages to be furnished with an economizer when an economizer is to be furnished (see 3.12).

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h. A flow switch shall be installed on the inlet feedwater piping and interlocked with the fuel-burning equipment to prevent start-up and initiate a unit shutdown if a no-flow condition is sensed.

3.11 Feedwater regulator. When furnished, steam boilers shall be provided with an automatic feedwater regulating system to maintain the boiler manufacturer's established steaming water level. The regulator shall be single element of duel element as specified (see 6.2). The regulating system shall maintain the setpoint within ±5 percent of the full range of allowable operating level. Feedwater regulators shall include the required sensing, actuating, regulating, and controlling equipment necessary to perform under the required conditions specified herein. The regulating system shall be provided complete with all necessary fittings, tubing, piping, mechanical linkages, wiring and other material or parts required to connect the operating components.

3.11.1 Feedwater regulating valve. Regulating valves shall be furnished with a double block and bleed piping bypass to allow the regulator valve to be removed while the boiler is operating. The regulating valve shall be equipped with a handwheel or knob for manual control and operation and shall have the manufacturer's standard position indicator. The valve shall be constructed with a balanced force design to allow the valve the move easily against the force of the feedwater flow. Valve seats shall be replaceable. Materials of construction shall be the manufacturer's standard for good corrosion resistance in boiler feedwater service.

3.11.2 Manual feedwater regulator control. When specified (see 6.2), the feedwater regulator system shall be furnished with a manual and automatic switch and a control mechanism for manually adjusting the feedwater regulator position. The direction in which the control mechanism is to be operated in order to raise or lower the water level shall be clearly indicated.

3.12 Economizer. When specified (see 6.2), a separately packaged economizer shall be furnished. The economizer shall be complete with prefabricated interconnecting breeching for attachment to the flue gas outlet and to the feed piping between the economizer and the feedwater inlet and shall require minimum field assembly. The economizer shall be equipped to heat feedwater. An automatic feedwater preheat system shall be provided when necessary to prevent flue gas condensation and potential corrosion. Temperature gages shall be furnished on the inlet and outlet piping of the flue gas and the feedwater. Manual valving also shall be provided to allow the feedwater to bypass around the economizer, however, there shall be no valve between the economizer tubes and the steam boiler or hot water generator relief valves to eliminate the need for a separate economizer relief valve. Economizers shall be furnished with soot blowers. The economizer shall be designed and constructed in accordance with the ASME Code. Materials used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions.

3.13 Soot blowers. The steam boiler or hot water generator shall be furnished with the necessary soot blowing equipment to remove soot accumulations from heat transfer surfaces. Soot blower shall be either steam or air operated as specified (see 6.2).

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3.13.1 Steam operated soot blowers. Steam operated soot blowers, when provided, shall be valve-in-head type and shall be furnished complete with wall sleeves, clamps, hangers, supports, operating chains and other appurtenances required for a complete installation. The blower elements shall be so arranged that all parts of the heating surfaces shall be cleaned of any soot deposits when rotated by manual or automatic means. Elements shall be of such length, diameter, and total nozzle area that for the operating pressure involved there will be no significant difference in the cleaning effect between the nozzle nearest the inlet and those farthest from the inlet of the element. Soot blower piping shall include a valve chain operated from the floor, in addition to an auxiliary steam-stop valve, and drain connection. Soot blowers shall be fitted with scavenging air ports and shall be connected by tubing to a draft source in a manner to provide sufficient air for continuous scavenging of the blowers while the steam boiler or hot water generator is firing during periods when soot blowers are not operated for cleaning purposes. Soot blowers shall be deactivated upon a boiler shutdown and require a manual reset.

3.13.2 Air operated soot blowers. Air operated soot blowers, when provided, shall be furnished complete with wall sleeves, clamps, hangers, supports, operating chains and other appurtenances required for a complete installation. The blower elements shall be so arranged that all parts of the heating surfaces shall be cleaned of any soot deposits when rotated by manual or automatic means. Elements shall be of such length, diameter, and total nozzle area that for the operating pressure involved there will be no significant difference in the cleaning effect between the nozzle nearest the inlet and those farthest from the inlet of the element. Soot blower piping shall include a valve chain operated from the floor and drain connection. Soot blowers shall be fitted with scavenging air ports and shall be connected by tubing to a draft source in a manner to provide sufficient air for continuous scavenging of the blowers while the steam boiler or hot water generator is firing during periods when soot blowers are not operated for cleaning purposes. Soot blowers shall be deactivated upon a boiler shutdown and require a manual reset.

3.14 Control systems. The basic programming and flame safeguard controls shall be furnished as an integrated system by the manufacturer and shall be designed to function in combination with other system components to ensure the functional integrity of the total system. Maximum use of common components for the combustion control system and limit interlocks shall be made to control oil, gas, and coal firing. Means shall be provided to start the furnace on natural gas or propane to bring the temperature of the bed up to a set point temperature which will assure combustion of unburned fuel and the switch over to the fuel to be fired. The control system shall be so arranged that in addition, oil can be burned simultaneously with coal at any percentage necessary or desired up to rated capacity of the generator. Each steam boiler or hot water generator shall be equipped with the following control systems:

a. Programming control systems: In conjunction with the electronic flame safeguard control, safety interlocks and limit controls, the programming control systems shall govern the pre-firing, light-off, and shutdown sequences.

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b. Combustion control system: This system shall automatically regulate the firing cycle in accordance with the load demand.

3.14.1 Programmed control system. The programmed control system shall be in accordance with the firing procedures of NFPA 85H.

3.14.2 Combustion safeguard system. The steam boiler or hot water generator shall be provided with a standard (nonproprietary) combustion safeguard system. The system shall have a repetitive self-checking circuit in conformance with 3.14.2.1. The system shall consist of a combustion safeguard element connected to the programming control system for shutdown of the burner in the event of ignition failure or flame failure during light off and operation. The system shall not be allowed to recycle after ignition failure or main flame failure without manual reset. The combustion safeguard system shall activate the fuel shut off valves in not greater than 4 seconds after a flame failure occurs. Combustion safeguard elements shall be installed so as to not respond to ignition spark, hot refractory, or the reflection of flames on atomizing media or oil spray. Combustion safeguard elements for igniter flames shall detect only flames of sufficient intensity to safely light the main burner. Flame sensing elements shall be so connected as to prevent initiation of start-up cycle if flame is present or if the safeguard system has failed in the flame sensing position. The combustion safeguard system shall be designed so that the failure of any component prevents restart after shutdown. The combustion safeguard system shall self-check automatically at the beginning of each startup cycle and not have a manual bypass of this function.

3.14.2.1 Repetitive self-checking circuit. The repetitive self-checking circuit shall automatically check all components of the combustion safeguard system including the flame detector. The component check of the detector is not required if the detector is fail-safe such as lead sulfide cells or flame rods. During operation the self-checking circuit shall test itself at intervals not greater than 6 seconds. Failure of any component to respond properly during this check shall cause an immediate shutdown of the steam boiler or hot water generator.

3.14.3 Safety interlocks and limit switches. Each programmed control shall include the safety interlocks and limit switches as defined and required in ASME CSD-1, NFPA 85A, and NFPA 85H as applicable. The safety interlocks and limit switches shall cause a recycling or nonrecycling safety shutdown as required in NFPA 85H. All nonrecycling shutdowns shall require a manual reset of the programmed control system. The safety interlocks and limit switches shall be in accordance with UL-353 or FM.

3.14.4 Combustion controls. Combustion controls shall automatically regulate the firing rate in accordance with the load demand and shall be capable of meeting the burner turndown ratio specified in 3.7.2. Control systems shall maintain air to fuel mixtures within the limits required for stable and efficient combustion throughout the specified firing range and, as applicable, during changes in firing rates. The combustion control systems shall maintain the steam pressure or water temperature as applicable, at the outlet connection. The combustion control shall release from the low-fire position and adjust the firing rate in accordance with the load demand except

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that release may be delayed for a period not greater than 60 seconds to ensure stabilization of the flame. On cold startup, release may be delayed on steam boilers or hot water generators equipped with controls which prevent movement to high fire until a predetermined water temperature is reached. A means shall be provided to permit manual control of the firing rate. However, the system shall be so designed that it shall be impossible to put fuel control on manual while the air control remains on automatic or vice versa. The systems shall function to cause a normal shut-down in response to a predetermined pressure for a steam boiler or water temperature for a hot water generator below the steam or relief valve setting when the load demand drops below the minimum firing rate provided by the control system, and results in a buildup of pressure or temperature to the predetermined set point. The combustion control system shall be capable of performing the following functions:

a. Means shall be provided to assure adequate primary air for the required fuel input to preclude the possibility of an unsafe fuel to air ratio condition.

b. The coal feed flow may be taken as the rate of feed.

c. The air to fuel mixture shall be maintained within safe limits as established by operation under any steam boiler or hot water generator output conditions within the controllable operating range of the system.

d. Means shall be provided for setting minimum and maximum limits on fuel and air control systems to prevent these systems from providing fuel and air flows beyond the stable flame limits of the fuel burning system. These minimum and maximum limits shall be defined by the manufacturer and verified by operating tests (see 4.7.2).

e. When changing the rate of furnace input, the air flow and fuel flow shall be changed simultaneously at the proper rates to maintain safe air to fuel ratio during and after the changes.

f. Maintain furnace balance draft (zero draft) at the desired set point in the furnace.

g. Means shall be provided to prevent the control system from demanding a fuel-rich mixture. Control action toward more fuel and less air shall be locked when the air to fuel ratio decreases below preset value.

h. Means shall be provided to cut back fuel to available air. If the air to fuel ratio decrease below an appropriate preset value, control action toward less fuel and more air shall be taken.

i. The combustion control system shall be interlocked with the coal fuel drive to balance the ash discharge with the firing rate.

j. Means shall be provided to permit as much on-line maintenance of combustion control equipment as possible.

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k. Means shall be provided for calibrations and check testing of combustion control and associated safeguard equipment.

3.14.5 Combustion control system. The combustion control system shall automatically regulate the firing cycle in accordance with the local demand. The combustion control system shall maintain air air to fuel mixture within the limits required for stable and efficient combustion throughout the required firing range and, as applicable, during changes in firing rates. The continuous maximum firing rate shall be as required to maintain the maximum boiler thermal capacity required in 3.1. The combustion control system shall maintain the load within 3 percent of the percent of the set point values of the operating control. The percentage tolerances shall not be applied in a manner that would permit the high-fire input of the burner to be increased beyond the maximum rated input marked on the boiler assembly.

3.14.5.1 Oxygen compensation system. When specified (see 6.1.1 and 6.2), an oxygen compensation system or trim shall be provided with the modulating combustion control systems specified in 3.14.5. The oxygen analyzer shall provide a feedback signal to the air to fuel ratio control system to maintain an adjustable preset percentage of excess oxygen in the flue gas. The controlling system shall be able to adjust the gain and offset of the feedback signal. Adequate safeguards shall be provided with the oxygen compensation control to prevent the oxygen compensation system from demanding a fuel rich mixture.

3.14.5.2 Oxygen compensation system with unburned combustible gas analyzer. When specified (see 6.1.1 and 6.2), the oxygen compensation system required in 3.14.5.1 shall be furnished with an unburned combustible gas analyzer. The unburned combustible gas analyzer shall measure the concentration of carbon monoxide or the concentration of hydrogen as specified (see 6.2), to determine the minimum level of oxygen required and transmit a corrective signal to the combustion control system. The unburned combustible gas analyzer shall be able to operate with oxygen present in the flue gas. The unburned combustible gas analyzer shall also be used to prevent the combustion control system from demanding a fuel rich mixture. An adjustable setpoint on the analyzer shall initiate an alarm and shall cause an immediate increase of the air to fuel ratio. The air to fuel ratio shall remain higher until the alarm condition no longer exists. The increase in the air demand signal during an alarm condition shall be not greater than 15 percent of the control position.

3.14.6 Bed temperature indicator. Temperature indicators shall be provided on the control panel to indicate the minimum and maximum primary bed temperature. The indicator shall be connected to the igniter controls so that when the primary bed is placed into operation all the igniters shall be lighted and shall extinguish the igniters only when the primary bed is at operating temperature. The indicator shall be also connected so that when the temperature of the primary bed drops below the minimum set temperature all the igniters shall be automatically restarted and sound an alarm and light a red indicating light. The indicator shall have acknowledge and reset buttons and be enclosed in a dust tight case suitable for flush mounting. When specified (see 6.2), the indicator shall record bed temperature. When the indicator is to record bed temperature, it shall be recorded on a strip or circular chart as specified (see 6.2).

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3.15 Ignition systems. Steam boilers and hot water generators shall be equipped with ignition systems designed to impinge on the main air to fuel mixture, and supply sufficient ignition energy to provide positive ignition of all fuel discharged to the fluidized bed under interlock light-off conditions. The igniter shall be integrated with the programming control systems specified herein. All igniters shall be of the intermittent type with gas-electric ignition. Ignition transformers shall be rated on the secondary side at not less than 6,000 volts for gas-electric igniters.

3.16 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems shall be mounted on a single control panel or cabinet insofar as practicable in order to centralize the control functions, and shall be designed for separate mounting. Switches shall include burner start, manual-automatic combustion control, fuel oil heater, and fuel selector switches, as applicable. Labeled indicating lights shall be provided and shall be activated when the interlock for the completion of the operation is activated in control sequences herein. The control panel or cabinet or the individual elements shall be provided with NEMA ICS 6, type 12 dust protective covers, and shall be mounted on a solid base and not be subjected to vibrations or be hazardous to the operator. All terminals requiring connection upon installation of the generator shall be permanently identified in such manner that correct connections can be easily made by reference to the instructions and wiring diagrams provided with the equipment. Terminal connections for control functions shall be segregated from power circuit terminals. Wiring shall be in accordance with NFPA 70. The location of the panel shall either be at the side of the steam boiler or hot water generator, or in a free-standing control panel for remote mounting as specified (see 6.2).

3.16.1 Indicating lights and alarm. The programming control system shall, as minimum, indicate the following occurrences with an indicating light on the control panel:

a. Activation of power on or start of blower motor.
b. Draft fan operation.
c. Pre-ignition purge cycle.
d. Trial for pilot ignition.
e. Detection of main flame.
f. Operating under automatic control.
g. Safety interlock or limit switch shutdown.
h. Low fuel oil supply pressure.
i. Low atomizing steam or air oil pressure.
j. High furnace pressure.
k. High furnace draft.
l. Low combustion air flow.
m. Loss of control system power.
n. Low bed temperature.

An audible alarm shall also be activated if a shutdown caused by a safety interlock or limit switch occurs. Additional indicating lights shall also be as specified (see 6.2).

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3.16.2 Annunciator. When specified (see 6.2), a separate annunciator shall be provided with the control panel. The decibel rating of the annunciator shall be as specified (see 6.2). Unless otherwise specified (see 6.2), the annunciator shall indicate all alarm conditions and nonrecycling burner shutdowns required in 3.16.1. The annunciator shall also have a manual reset button and a capability to indicate the first cause of a boiler shutdown. The annunciator shall not require special or additional limit devices, or change the sensitivity of limit switches, or annunciate without the limit devices being actuated.

3.17 Fluidized bed coal burner. A class 6, class 7, class 8 or class 9 steam boiler or hot water generator shall be furnished with a fluidized bed coal burner. The fluidized bed coal burner shall include a coal hopper-feeder, limestone hopper-feeder, fuel and air chamber, fluidized bed burner, air nozzles, ducts and piping with drive and damper, valves and actuators for the combustion process turndown (see 3.7.2). The burner shall be properly sized and interconnected to satisfy the functional requirements and not interfere with the combustion process. The fuel burning system shall provide means for safe start-up, operation and shut-down of the combustion process. This shall include appropriate openings and configurations in the component assemblies to permit suitable observations, measurement, and control of the combustion process. The fuel burning system design shall be based on the following requirements:

a. The air supply equipment for combustion shall be properly sized, arranged, and supervised to ensure a continuous, steady air flow adequate for all operating conditions.

b. The air inlets, duct work, shall minimize contamination of the air supply by such materials as flue gas, water, and fuel.

c. The air equipment shall supply proper air flow during and furnace pressure pulsations.

d. Means shall be provided to control the feeder outlet temperature within limits suitable for coal and limestone being fired.

e. Piping which connects the feeder systems to fluidized bed shall be kept to a minimum length, sized and arranged to minimize unequal distribution of the coal and limestone.

f. Fuel (coal and limestone) shall be supplied to the furnace continuously and within stable flame limits, independent of the auxiliary ignition subsystems over the operating range of the fuel burning equipment.

g. Means shall be provided for adequate observation of the conditions at each burner subsystem and its ignition zone space shall be provided for flame safeguard control equipment.

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h. Means shall be provided for rodding or cleaning of the coal and limestone feeders if coking of the coal shall occur.

i. Convenient access for maintenance and housekeeping shall be provided.

3.17.1 Coal hopper-feeder. The coal hopper shall be properly sized and arranged to ensure a continuous, steady flow adequate for all operating requirements without bridging of the fuel. The hopper shall include a variable speed feeder to assure a turndown as specified in 3.7.2 for introduction of the coal into the furnace. The feeder shall be designed with sufficient capacity range to allow for variations in size, quality, and moisture content of the coal as specified in 3.7.1.1 to hold not less than one hour firing at maximum thermal output capacity. Each feeder drive shall be protected with safety release devices such as shear pins or stall clutches shall be provided to prevent damage to the coal feed device due to foreign materials in the coal.

3.17.2 Limestone hopper-feeder. The limestone hopper shall be properly sized and arranged to ensure a continuous steady flow adequate for all operating requirements without bridging of the crushed stone. The hopper shall include a variable speed feeder to adjust the volume of limestone being introduced to compensate for the varying sulphur content of the coal being fired. The capacity of the hopper shall be sufficient to sustain adequate sulfur removal for one hour at maximum thermal output capacity.

3.18 Fluidized bed oil burner. Class 1, class 2, class 4, class 5, class 7, or class 9 steam boilers or hot water generators shall be furnished with a fluidized bed oil burner. The burner shall produce a clean, stable flame at all firing rates using the grade of fuel oil specified in 3.7.1.2. The burner flame shape shall fit the furnace without impingement on or overheating of the furnace walls. Burner parts shall be protected from overheating or fabricated from materials sufficient to withstand the temperatures they are exposed to. The design of the fuel oil system including safety interlocks, limit switches and control valves shall comply with the applicable requirements of ASME CSD-1, NFPA 85A and NFPA 85H. The fuel oil piping shall be arranged in accordance with one of the recommended piping layout options contained in the appendices of NFPA 85A and NFPA 85H. Fuel oil piping shall withstand 1.5 times the maximum operating pressure of the oil supply system without leakage. Oil burners of the return flow type shall be equipped with a return flow safety shutoff and a spring-loaded check valve to prevent return oil leakage on safety shutdown. The fuel system may operate on either the constant pressure or the constant differential principle provided positive control and adequate atomizing pressure is maintained at all firing rates. A fuel oil temperature gage shall be provided. The fuel oil system include fuel oil pump, strainer, relief valve and trim heater as required below.

3.18.1 Fuel oil pump. The fuel oil pump shall be of the motor-driven positive displacement or variable displacement metering type. The pump shall deliver the required fuel oil with the oil supply temperature and pressure conditions specified (see 3.7.1.2) to the burner at sufficient quantity and pressure to maintain stable firing conditions at the maximum required firing rate.

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3.18.2 Strainer. A fuel oil strainer shall be furnished to assure the delivery of clean oil to the burner and prevent plugging of the interconnecting piping and valves. The strainer shall be located in front of the pump so that oil supplied to the boiler passes through the strainer first. A piping bypass shall be furnished to allow the strainer to be cleaned without shutting down the oil flow to the burner.

3.18.3 Relief valve(s). An automatic oil pressure relief valve(s) shall be provided on the discharge side of the fuel oil pump, the discharge side of the fuel oil heaters, and such other locations as may be necessary to prevent excess pressure to the burner. The relief valve(s) shall be complete with return piping to the fuel oil suction return line.

3.18.4 Trim heater. When specified (see 6.2), an auxiliary electric trim heater shall be provided to reduce the fuel oil viscosity prior to atomization. The trim heater shall be designed to heat 80 percent of the maximum fuel oil flow from the specified oil supply temperature to the required atomization temperature. The trim heater and associated controls shall be located on the packaged unit assembly. When the required fuel oil temperature is reached and maintained by the heater, a thermostatic control or limit switch shall turn off the heater. Trim heater operating voltage shall not be greater than 480 volts. When provided, electric trim heaters rated up to 1800 Watts (W) shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. The electric power and separate control circuits shall be protected by circuit breakers located on the packaged unit assembly. An outlet temperature gage shall be provided.

3.19 Fluidized bed gas burner. Class 3, class 4, class 5, class 8 or class 9 steam boilers or hot water generators shall be furnished with a fluidized bed gas burner. The burner shall produce a clean, stable flame at all firing rates using the gas fuel specified in 3.7.1.2. The burner flame shape shall fit the furnace without impingement on or overheating of the furnace walls. Burner parts shall be protected from overheating or fabricated from materials sufficient to withstand the temperatures they are exposed to. The design of the system including safety interlocks, limit switches and control valves shall comply with the applicable requirements of ASME CSD-1, NFPA 85A, and NFPA 85H. The fuel gas piping shall be arranged in accordance with one of the recommended piping layout options contained in the appendices of NFPA 85A and NFPA 85H. Fuel gas piping shall withstand 1.5 times the maximum operating pressure of the fuel gas supply system without leakage. The design shall incorporate an effective means of preventing flame propagation back into the gas or supply lines.

3.20 Fuel oil preheaters. When specified (see 6.1.1 and 6.2), fuel preheating systems shall be furnished as part of the packaged unit in place of the trim preheater specified in 3.18.4. The heater shall be either electric or steam as specified (see 6.2). The preheater and associated controls shall be located on the packaged unit assembly. The preheating equipment shall be designed in accordance with Section VIII of the ASME Code. The system shall be designed to heat 120 percent of the maximum fuel oil flow from the specified oil supply temperature (see 3.7.1.2 and 6.2) to the required atomization temperature. Pressure drop across the preheater shall be not greater than 15 psi. When the required fuel oil temperature is reached and maintained by the

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heater, a thermostatic control or limit switch shall turn off the heater. Temperature gages shall also be provided with a manual reset. Temperature gages shall be provided to measure fuel oil inlet and outlet temperatures across the preheater.

3.20.1 Electric preheater. When the electric preheater is required (see 3.20), the preheater shall be furnished with a temperature controller to maintain the fuel oil temperature within 5 percent of the set point temperature. Preheater operating voltage shall not be greater than 460 volts. Electric preheaters rated up to 1800 W shall be controlled by a line voltage thermostat switch or by a magnetic contactor with a coil operated by a thermostat. Preheaters rated at greater than 1800 W shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. The electric power and separate control circuits shall be protected by a circuit breaker located on the boiler assembly.

3.20.2 Steam preheater. When the steam preheater is required (see 3.20), the heater shall be designed for the steam boiler design pressure specified in 3.7 and shall use steam generated by the steam boiler. The preheater shall be furnished with an adjustable oil temperature regulator to maintain the fuel oil temperature with 5 percent of the set point temperature. Means shall be provided for visually detecting oil in the preheater condensate line.

3.20.2.1 Electric preheater within steam preheater. When specified (see 6.2), the steam preheater shall be furnished with an electric preheater in the same exchanger shell for cold start up. The electric preheater shall have sufficient capacity for the firing rate occurring when the combustion controls are in the position for low fire starts. The electric preheater shall be furnished in accordance with the requirements of 3.20.1.

3.21 Electrical requirements. Electrical wiring design practices shall be installed in accordance with the applicable requirements of NFPA 70. The steam boiler or hot water generator and associated equipment shall obtain power from a circuit with a single master manual service disconnect switch and automatic circuit breaker for all steam boiler or hot water generator electrical equipment. Motors shall be in accordance with the provisions of NEMA MG 1, high efficiency, shall be totally enclosed fan cooled and shall be designed for operation on the voltage, phase, and frequency as specified (see 6.2). Unless otherwise specified (6.2), motors of 100 horsepower and less shall be provided with magnetic across-the-line starter and overload protection. Motors over 100 horsepower shall have reduced voltage starting. Motor starters shall conform to the applicable requirements of NEMA ICS 1 and ICS 2. Motor starters and controls shall be enclosed in NEMA ICS 6 Type 12 enclosures and may be installed in the control panel of 3.16.

3.24 Cleaning, treatment, and painting. Bare steel surfaces shall be cleaned and a primer coat applied. Paint shall be applied to surfaces normally painted in good commercial practice.

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3.25 Lifting and tiedown attachments. When specified (see 6.2), the steam boiler or hot water generator shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the unit on the carrier when shipped.

3.26 Mounted operating instructions. Unless otherwise specified (see 6.2), operating instructions shall be summarized on a decal or corrosion resistant plate to be mounted inside control panel door or similar weatherproofed location.

3.27 Identification marking. Identification shall be permanently and legible marked directly on the steam boiler of hot water generator or on a corrosion-resistant metal plate. Identification shall include manufacturer's name, unit model and serial number, thermal capacity, ASME code stamp, year of manufacture and other pertinent information as identified in the ASME Codes. The Department of Defense contract number that this equipment is procured under shall also be marked on the equipment or identification plate.

3.28 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each boiler. When furnished, the spare parts required and the quantity thereof shall be specified (see 6.2).

3.29 Factory start-up service. Unless otherwise specified (see 6.1.1 and 6.2), factory authorized service personnel shall be provided to supervise installation and start-up of the steam boiler or hot water generator for two consecutive weeks consisting of five 8-hour working days per week.

3.30 Workmanship. Workmanship shall be of such quality as to produce steam boilers or hot water generators in accordance with the ASME Code, as applicable, and shall meet the requirements of this specification and standards prevailing among manufacturer's who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance. This inspection shall be performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the quality conformance inspection shall include the examination of 4.3, the pressure test of 4.4, the relief valve test of 4.5, when the inner casing is part of the manufacturer's standard design, the test of 4.6, and the packaging inspection of 4.8.

4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the on-site inspection shall consist of the tests of 4.7. When specified (see 6.2), the examination of 4.3 and the tests of 4.4, 4.5, and when the inner casing is part of the manufacturer's standard design, the test of 4.6 shall be repeated on-site.

4.3 Examination. Each steam boiler or hot water generator supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability.

4.4 Fuel system pressure test. The preassembled fuel system piping furnished integrally with the burner shall be checked for leakage. The fuel system piping shall be hydrostatically tested for not less than 30 minutes at 1.5 times the maximum operating pressure without leakage.

4.5 Relief valve set pressure test. Relief valves required by the ASME Code shall be individually tested on a test bench using a gas to confirm the required set pressure. The test shall be conducted at a facility that normally performs this work.

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4.6 Inner casing test. When an inner casing is included in the manufacturer's standard design, the furnaces and flue-gas passages of boilers normally operated under pressure shall be tested for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches of water, whichever is less. The test shall be performed prior to the installation of the insulation and outer casing. The section of the unit to be tested shall be carefully sealed using suitable means for blanking any area not to be subjected to test pressure. Air shall be admitted to the unit until the test pressure is reached and then the unit shall be carefully sealed. The outer casing shall be vented to the atmosphere during the casing pressure test. If the pressure does not drop greater than 5 inches of water in ten minutes, the unit shall be regarded as satisfactorily tight and accepted. Failure to meet the requirements of 3.9.14 shall constitute cause for rejection. Any change in air temperature which affects the test conditions shall be taken into consideration.

4.7 On-site tests. Operational fire tests and auxiliary equipment tests shall be performed on each installed unit. These tests shall be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the boiler manufacturer is accepted at the discretion of the contracting officer or authorized representative. Acceptable fire test results from identical or similar boiler equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured with this specification is in compliance with one or more of the tests of 4.7.1 through 4.7.9.

4.7.1 Operational tests. The steam boiler or hot water generator shall be tested to demonstrate control and operational conformance to the requirements of this specification under varying load conditions ranging from the specified maximum capacity to the minimum turndown firing rate. The unit shall be tested for ability to make a cold start. The transfer of control from automatic to manual and from manual to automatic shall be performed on the overall system and independently on each automatic and manual subsystem. This transfer shall be smooth without the need for special adjustments to trim the combustion control devices. The air to fuel ratio controller shall be tested by attempting to create a fuel rich operating condition below the manufacturer's pre-set limit. Steam boilers or hot water generators that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.

4.7.2 Capacity and efficiency tests. The capacity, efficiency, and if required, steam purity at the specified maximum capacity of the steam boiler shall be determined in accordance with the ASME Performance Test Code PTC 4.1. The ASME Test Form for Abbreviated Efficiency test may be used. The efficiency shall be determined by the heat loss method. Performance shall be as specified in 3.8. Methods of measuring the flow of steam, water, and fuel other than weighing will be allowed. The accuracy of the method shall assure the Government the desired performance specified in 3.8 and be acceptable to the contracting officer or his authorized representatives. Failure to meet the requirements of 3.8 shall constitute cause for rejection.

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4.7.3 Equipment tests. Control panel, preheaters, insulation, valves, controllers, regulators, soot blowers, electric motors, dampers, furnace access provisions, and other related equipment shall be operated as part of the capacity and efficiency tests. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth without backlash.

4.7.4 Sequencing. The steam boiler or hot water generator shall start, operate, and stop in strict accordance with the specified operating sequence with the specified fuel systems.

4.7.5 Limit and cutout switches. Safety shutdowns shall be caused by simulating fuel safety interlock actuating conditions for each limit and cutout switch, specified in 3.10. The safety shutdowns shall occur without damage to the equipment.

4.7.6 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.7.7 Hot refractory test of combustion safeguard system. The installed burner shall be operated at high fire until the furnace attains steady state condition. The main fuel valve shall then be closed manually. The combustion safeguard shall drop out causing the safety shutoff valves to close within the specified flame-failure reaction time.

4.7.8 Flame failure. The operation of the flame failure control shall be verified by simulated flame and ignition failures. The igniter establishing period, the main-burner establishing period, flame-failure reaction time, and the safety-shutoff valve closing times shall be verified by stop watch.

4.7.9 Emissions testing. The required emissions testing for opacity, particulates, nitrogen oxides, sulfur dioxide, and carbon monoxide shall be conducted in accordance with the procedures of the governing Federal, state, or local environmental agency as specified (see 6.2).

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 4 of MIL-B-3180.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3180 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The steam boilers and hot water generators covered by this specification are intended for generation of steam or hot water to be used in processing operations and space heating systems in land installations.

6.1.1 Selection factors. The following should be considered in the selection of a steam boiler or hot water generator and the exercise of options herein:

a. The steam boilers or hot water generators with mechanical collectors alone will probably not meet existing particulate emission standards, without additional emission control equipment.

b. The class of steam boiler or hot water generator (oil, gas, coal, or combination) should be determined in accordance with the availability or economics of the fuel supply and any DoD requirements for capability of firing multiple fuels (see 3.1).

c. Normally, oil-fired steam boilers or hot water generators above 5,000,000 Btu/hr output should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6. When changing grade of fuel, replacement may be necessary (1) of the fuel nozzles because of the different fuel viscosity, and (2) of the fuel pump motor to drive the pump at a speed that ensures required fuel pressure in order to maintain the required burner output and meet the optimum air to fuel ratio (see 3.1).

d. The steam boiler or high temperature hot water generator should be sized to permit efficient operation during the summer when low heating requirements may occur (see 3.7.2).

e. Burner turndown ratios higher than those required or useful should not be specified. Steam boilers or hot water generators are usually limited to a minimum firing rate of 33.3 percent of maximum firing rate with an option minimum firing rate of 16.7 percent of the maximum firing rate with oxygen compensation, (see 3.7.2).

f. If the steam boiler or hot water generator is subject to environmental regulations, each agency should consult with their environmental support office since regulation limits of particulate and emissions for they vary from city, state, and Federal agencies (see 3.8.6).

g. Hot water generator systems should use a nitrogen pressured vessel for pressure control. The pressure control vessel is not part of the typical packaged steam boiler or hot water generator. The feedwater controller for the hot water generator should be mounted on the nitrogen pressured vessel (see 3.9.1).

h. The cost of flue gas analyzers for oxygen should be justified based on the anticipated fuel savings they should produce. They tend to be more economic on larger steam boilers or hot water generators with constantly changing loads and fuel quality (see 3.14.5.1 and 3.14.5.2).

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i. A packaged steam boiler or hot water generator mounted preheater should be specified only at installations of single heavy oil fired steam boilers or hot water generators. It is more economic for multiple steam boiler or hot water generator installations to use a common preheater that is external from the individual packaged steam boiler or hot water generator skids (see 3.20).

j. Factory authorized service should be specified only if the steam boiler or hot water generator will be installed and tested without a long period of interim storage (see 3.29).

k. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the boiler is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

l. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits for Class C3, Group I equipments as specified in MIL-STD-461. The boiler should be subjected to tests to determine conformance in accordance with MIL-STD-462.

6.2 Acquisition documents. Acquisitions documents should specify the following:

a. Title, date, and number of this specification.
b. Type, size, and class required (see 1.2 and 3.1).
c. Issue of DoDISS to be cited in the soliciation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Required thermal output capacity (see 3.1).
e. Design parameters of the steam boiler or hot water generator (see 3.7).
f. Type of fuel for firing the steam boiler or hot water generator. Whether the start up fuel is to be either natural gas or propane (see 3.7.1).
g. Coal characteristics (see 3.7.1.1).
h. Fuel characteristics, if other than as specified (see 3.7.1.2).
i. Grade of fuel oil, fuel oil supply temperature, and fuel oil supply pressure, when oil burners are required (see 3.7.1.2 and 3.20).
j. Fuel gas characteristics for gas burners, if other than as specified (see 3.7.1.2).
k. Continuous stable low firing rate, if other than as specified (see 3.7.2).
l. When the flue gas exit temperature may be less than 325°F (see 3.8.4).
m. Emission requirements for opacity, particulates, nitrogen oxides, sulfur dioxide, and carbon monoxide (see 3.8.6).

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n. When the manufacturer is to provide the following information for use in the design and installation of a flue gas baghouse (see 3.8.7):

(1) Maximum gas flow in standard cubic feet per minute.
(2) Required draft at boiler flue gas outlet in inches of water at maximum gas flow conditions.
(3) Dimensions of flue gas outlet.
(4) Height of flue gas outlet above grade and relative orientation.

o. Water pressure drop, if other than as specified, when a hot water generator is provided (see 3.9.2).
p. Design of combustor bed, if other than as specified (see 3.9.3).
q. Type of ash removal system required (see 3.9.11).
r. Trim and appurtenances, if other than as specified (see 3.10).
s. Whether the feedwater regulator is to be single element or double element (see 3.11).
t. When the feedwater regulator is to be provided with both a manual and automatic switch and a control mechanism for manually adjusting the feedwater regulator (see 3.11.2).
u. When a separately packaged economizer is to be furnished (see 3.12).
v. Whether the soot blower is to be either steam or air operated (see 3.13).
w. When oxygen compensation is to be provided with the modulating combustion control system (see 3.14.5.1).
x. When the oxygen compensation system, if provided, is to be furnished with an unburned combustible gas analyzer (see 3.14.5.2).
y. The concentration of carbon monoxide or hydrogen the unburned combustion gas analyzer is to measure (see 3.14.5.2).
z. When the bed temperature indicator is to record bed temperature and, if recording is required, whether it is on a strip or circular chart (see 3.14.6).
aa. Whether the control panel is to be either at the side of the steam boiler or hot water generator, or be in a free-standing control panel for remote mounting (see 3.16).
bb. When additional indicating lights are required, and when required, what the additional lights are to indicate (see 3.16.1).
cc. Decibel rating of the annunciator (see 3.16.2).
dd. Indications of annunciator, if other than as specified (see 3.16.2).
ee. When an auxiliary electric trim heater is required (see 3.18.4).
ff. When fuel preheating systems are to be provided in lieu of the trim heater and, if provided, whether the preheating equipment is to be electric or steam (see 3.20).
gg. When the steam preheater is to be furnished with an electric preheater for cold start ups (see 3.20.2.1).
hh. Power, voltage, phase, and frequency for motor operation design (see 3.21).
ii. When lifting and tiedown attachments are required (see 3.25).
jj. Mounted operating instructions, if other than as specified (see 3.26).

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kk. When spare parts are required and, if required, the required spare parts and quantity (see 3.28).
ll. Provisions for factory authorized service personnel, if other than as specified (see 3.29).
mm. Quality conformance inspection, if other than as specified (see 4.2.1).
nn. On-site inspection, if other than as specified (see 4.2.2).
oo. When the examination of 4.3, fuel pressure test of 4.4, the relief valve set pressure test of 4.5, and the inner casing test of 4.6 are to be repeated at the on-site inspection (see 4.2.2).
pp. Testing procedures for opacity, particulates, nitrogen oxides, sulfur dioxide and carbon monoxide (see 4.7.9).
qq. Required level of preservation and level of preservation (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Reference Paragraph	DID Number	DID Title
6.3.1.1, 6.3.1.2, 6.3.1.3, 6.3.1.4, 6.3.1.5, 6.3.4 and 6.3.5	MISC-80678	Certification Data/Report
6.3.2	DI-S-3607	Schematic Logic Diagram
6.3.3	UDI-E-21344	Instruction, Equipment Installation
6.3.6	DI-M-2050	Commercial Manuals

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3.1 Compliance. Prior to approval of shipment, the contractor should submit to the contracting officer or authorized representative satisfactory evidence that the proposed steam boiler or hot water generator to be furnished under this specification meets the requirements of Section I, Section IV, and Section VIII of the ASME Code, ASME CSD-1, UL 353, NFPA 70, NFPA 85A, and NFPA 85H.

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6.3.1.1 AMSE Code requirements. Acceptable evidence of meeting the applicable requirements of Section I, Section IV, and when an economizer or preheater is required, Section VIII of the ASME Code should be the ASME Code symbol and when specified, receipt of the completed manufacturer's data reports for the specified steam boiler, hot water generator, or pressure vessel

6.3.1.2 ASME CSD-1 requirements. Acceptable evidence of meeting the applicable requirements of ASME CSD-1 should be the receipt of a manufacturer's certificate of compliance, stating the equipment procured with this specification is in accordance with the applicable requirements of ASME CSD-1.

6.3.1.3 NFPA requirements. Acceptable evidence of meeting the applicable requirements of NFPA 70, NFPA 85A, and NFPA 85H should be the receipt of a manufacturer's certificate of compliance stating the equipment procured with this specification is in accordance with the applicable requirements of NFPA 70, NFPA 85A, and NFPA 85H.

6.3.1.4 UL requirements. Acceptable evidence of meeting the applicable requirements of UL 353 should be the certification symbol, or listing mark on the products, or a certified test report, from a recognized independent testing laboratory indicating the boiler controls or burner have been tested and conform to UL requirements.

6.3.1.5 FM requirements. Acceptable evidence of meeting the applicable requirements of the FM Approval Guide should be an FM listing or certification symbol or label, or a certified test report from a recognized independent testing laboratory indicating that the components have been tested and conform to FM requirements.

6.3.2 Schematic Logic Diagram. The manufacturer should furnish a schematic logic diagram of the combustion control system (see 3.14.4).

6.3.3 Remote control panel installation. When a free-standing remote mounted control panel is to be furnished (see 3.16), installation instructions covering the connection of the control panel should be provided.

6.3.4 Design verification. The manufacturer should furnish data to verify compliance with the specified thermal heat input rates (see 3.8.3).

6.3.5 Relief valve. A certificate of compliance verifying the performance and results of the relief valve pressure test (see 4.5) should be provided to the contracting officer.

6.3.6 Technical manuals. The manufacturer should provide two commercial technical manuals normally prepared and supplied with the boiler.

6.3.7 Boiler operational certification (Navy Only). The boiler operational certification should be performed, where applicable, per NAVFAC MO-324, after the boiler complies with the on-site tests (see 4.7). The boiler inspector should be certified by NAVFAC and licensed by the Engineering Field Division.

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6.4 Definitions. Definitions used in this specification have been in accordance with the ABMA Lexicon. Other terms used in this specification should be interpreted in accordance with the referenced national standards.

6.4.1 Effective radiant heating surface. Effective radiant heating surface is defined as the heat exchange surface directly exposed to the radiant heat of the flame on one side and to the water or water-steam mixture being heated on the other. Effective heating surface should be calculated on the side receiving heat and should neglect any extended surfaces.

6.4.2 Furnace volume. Furnace volume should be expressed in cubic feet and should be the product of the interior cross-sectional area and height of the furnace. The furnace should be considered as not extending beyond the inlet of the convection section.

6.4.3 Total generator heating surface. The total generator heating surface should be expressed in square feet and should consist of those surfaces which are exposed to the products of combustion on one side and water on the other. The total generator heating surface should be measured on the side receiving the heat.

6.5 Cross-reference of classifications. The classification of packaged fluidized bed units in this document differs from the previous revision in the following respects:

MIL-B-29205	MIL-B-29205A
TYPES	TYPES
Type CLO - Combination coal-fired and light-oil-fired (grade 2 or 4, as specified).	Type CLO - Deleted.
Type CHO - Combination coal-fired and heavy-oil-fired (grade 5 or 6, as specified).	Type CHO - Deleted.
Type CNG - Combination coal-fired and natural-gas-fired.	Type CNG - Deleted.
Type CGL - Combination coal-fired, gas-fired, and light-oil-fired (type of gas and grade of oil as specified).	Type CGL - Deleted.
Type CGH - Combination coal-fired, gas-fired, and heavy-oil-fired (type of gas and grade of oil as specified).	Type CGG - Deleted.

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MIL-B-29205 - Continued

TYPES - (Continued)

Type I - None

Type II - None

Type III - None

Type IV - None

Type V - None

Type VI - None

SIZES

Size 1 - 2,000,000 to 2,500,000 Btu/hr.

Size 2 - 2,500,001 to 5,000,000 Btu/hr.

Size 3 - 5,000,001 to 10,000,000 Btu/hr.

Size 4 - 10,000,001 to 20,000,000 Btu/hr.

MIL-B-29205A - Continued

TYPES - (Continued)

Type I - Steam boiler producing low pressure steam not greater than 15 psig.

Type II - Steam boiler producing high pressure steam between 16 and 250 psig.

Type III - Steam boiler producing high pressure steam between 251 and 350 psig.

Type IV - Steam boiler producing high pressure steam between 351 and 600 psig.

Type V - Hot water generator producing low temperature water not greater than 160 psig or 250°F.

Type VI - Hot water generator producing high temperature water not less than 160 psig or 250°F.

SIZES

Size 1 - Deleted.

Size 2 - Boiler output capacity between 330 and 10,300 PPH from and at 212°F and atmospheric pressure at sea level.

Size 3 - Boiler output capacity between 10,301 and 36,100 PPH from and at 212°F and atmospheric pressure at sea level.

Size 4 - Boiler output capacity between 36,101 and 51,500 PPH from and at 212°F and atmospheric pressure at sea level.

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MIL-B-29205 - Continued	MIL-B-29205A - Continued
SIZES - (Continued)	SIZES - (Continued)
Size 5 - 20,000,001 to 30,000,000 Btu/hr.	Size 5 - Deleted.
Size 6 - 30,000,001 to 40,000,000 Btu/hr.	Size 6 - Deleted.
Size 7 - 40,000,001 to 50,000,000 Btu/hr.	Size 7 - Deleted.
CLASSES	CLASSES
Class 1 - None.	Class 1 - Heavy oil fired.
Class 2 - None.	Class 2 - Light oil fired.
Class 3 - None.	Class 3 - Gas fired.
Class 4 - None.	Class 4 - Combination heavy oil and gas fired.
Class 5 - None.	Class 5 - Combination light oil and gas fired.
Class 6 - None.	Class 6 - Coal fired.
Class 7 - None.	Class 7 - Coal/oil fired.
Class 8 - None.	Class 8 - Coal/gas fired.
Class 9 - None.	Class 9 - Coal/oil/gas fired.

6.6 Subject term (key word) listing.

Coal fired
Gas and oil fired
High and low pressure
Hot water generator

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Navy - YD

(Project 4410-0075)

Review activities:
Army - CE
Air Force - 84

User activities:
Army - CE
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-29205A	2. DOCUMENT DATE (YYMMDD) 90/09/17

3. DOCUMENT TITLE: BOILERS, FLUIDIZED BED, STEAM AND HOT WATER, PACKAGED TYPE (330 TO 51,500 POUNDS PER HOUR THERMAL OUTPUT CAPACITY)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

POC: 1562G

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME D. C. MUI, Project Manager Facilities Criteria Development Division	b. TELEPHONE (Include Area Code) (1) Commercial: (805) 982-5666 (2) AUTOVON: 551-5666
c. ADDRESS (Include Zip Code) Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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